

Analysis Tutorial 4

Problem 10

(a) ① Since $M \in \mathcal{A}$, $M \in M \cap \mathcal{A}$.

② Let $X \in M \cap \mathcal{A}$, then $X = M \cap A$ for some $A \in \mathcal{A}$. Then $\bar{X} = M \setminus X = M \cap (\overline{M \cap A})$
$$= M \cap (\bar{M} \cup \bar{A})$$
$$= (M \cap \bar{M}) \cup (M \cap \bar{A})$$
$$= M \cap \bar{A}$$

④ But $\bar{A} \in \mathcal{A}$ so $\bar{X} \in M \cap \mathcal{A}$.

③ Let $\{X_n\}_{n \in \mathbb{N}} \subseteq M \cap \mathcal{A}$. For each $n \in \mathbb{N}$, there exists $A_n \in \mathcal{A}$ such that

$$X_n = M \cap A_n.$$

$$\text{Then } \bigcup_{n=1}^{\infty} X_n = \bigcup_{n=1}^{\infty} M \cap A_n = M \cap \underbrace{\left(\bigcup_{n=1}^{\infty} A_n \right)}_{\in \mathcal{A}} \in M \cap \mathcal{A}$$

So $M \cap \mathcal{A}$ is a σ -algebra on M .

(b) We wish to show $X = \{A \in \mathcal{A} : A \in M\} = \{M \cap A : A \in \mathcal{A}\} \stackrel{Y}{=}$
Let $A \in X$. Then $A \in \mathcal{A}$ and $A \in M$. So

① $A = M \cap A \Rightarrow A \in Y$. Now let $B \in Y$. Then

$B = M \cap A$ for $A \in \mathcal{A}$. But $M \cap A \subseteq M$ and $M \in \mathcal{A} \Rightarrow B \in \mathcal{A}$ so $B \in X$.

So $X = Y$.

(c) ① We have shown $M \cap \mathcal{A}$ is a σ -algebra on M .

② $M_{M \cap \mathcal{A}}(\emptyset) = M(\emptyset) = 0$ since M is a measure

③ Let $\{X_n\} \subseteq M \cap \mathcal{A}$ be p.d. sets.

Then $\{X_n\} \subseteq \mathcal{A}$.

$$\text{So } M_{M \cap \mathcal{A}}\left(\sum_{n=1}^{\infty} X_n\right) = M\left(\sum_{n=1}^{\infty} X_n\right) = \sum_{n=1}^{\infty} M(X_n)$$
$$= \sum_{n=1}^{\infty} M_{M \cap \mathcal{A}}(X_n)$$

So $M_{M \cap \mathcal{A}}$ is a measure

Problem 11

(a) (i) Let $M \in \mathcal{A}$ be negligible and $X \in \mathcal{M}$.
Then there exists $N \in \mathcal{A}$ with $M \subseteq N$
and $\mu(N) = 0$. But $X \in \mathcal{M}$ implies X
is negligible.

(ii) Let $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$ be null.

Let $B_1 := A_1$, $B_n := A_n \setminus A_{n-1}$.

Since $B_n \subseteq A_n$, $\mu(B_n) \leq \mu(A_n) = 0$.

So $\mu(B_n) = 0$. Also $\{B_n\}_{n=1}^{\infty}$ is p.d.

So $\mu(\bigcup_{n=1}^{\infty} A_n) = \mu(\sum_{n=1}^{\infty} B_n) = \sum_{n=1}^{\infty} \mu(B_n) = 0$.

So $\bigcup_{n=1}^{\infty} A_n$ is null.

Now let $\{A_n\}_{n=1}^{\infty} \subseteq \Omega$ be negligible.

For any $n \in \mathbb{N}$, there exists $N_n \in \mathcal{A}$ such
that $A_n \subseteq N_n$ and $\mu(N_n) = 0$.

But then $\bigcup_{n=1}^{\infty} A_n \subseteq \bigcup_{n=1}^{\infty} N_n$ with $\mu(\bigcup_{n=1}^{\infty} N_n) = 0$
by above. So $\bigcup_{n=1}^{\infty} A_n$ is negligible.

(iii) Let $\Omega = \{1, 2, 3\}$

$\mathcal{A} = \{\emptyset, \Omega\}$

$\mu(A) = 0$ for all $A \in \mathcal{A}$

Then $\{1\} \subseteq \{1, 2, 3\} \in \mathcal{A}$ and $\mu(\{1, 2, 3\}) = 0$
but $\{1\} \notin \mathcal{A}$

(b) (i) Let $A \subseteq \mathbb{R}$ be countable. Then

$A = \bigcup_{n=1}^{\infty} \{a_n\}$, and $\lambda(A) = \lambda(\bigcup_{n=1}^{\infty} \{a_n\}) = \sum_{n=1}^{\infty} \lambda(\{a_n\})$
 $= 0$

(taking each a_n distinct)

(ii) $\mathbb{Q}$ is a countable subset of $\mathbb{R}$, so

$$\lambda(\mathbb{Q}) = 0.$$

$$\text{Since } \mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}), \lambda(\mathbb{R}) = \lambda(\mathbb{Q}) + \lambda(\mathbb{R} \setminus \mathbb{Q}) \\ = \lambda(\mathbb{R} \setminus \mathbb{Q})$$

$$\text{and } \lambda(\mathbb{R}) = \infty \text{ so } \lambda(\mathbb{R} \setminus \mathbb{Q}) = \infty$$

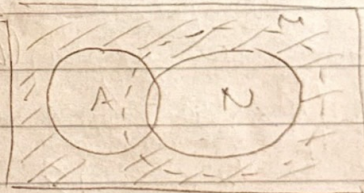
Problem 12

(a) ① $\Omega \in \mathcal{A}$ and $\emptyset \in \mathcal{N}$ so $\Omega \cup \emptyset = \Omega \in \tilde{\mathcal{A}}$

② Let $A \in \tilde{\mathcal{A}}$, then $A = A' \cup N$ for $A' \in \mathcal{A}$ and $N \in \mathcal{N}$

Then there is some $M \in \mathcal{A}$ with $N \subseteq M$ and $\mu(M) = 0$

$$\begin{aligned} \bar{A} &= \overline{A' \cup N} \\ &= (\underbrace{\bar{A'} \cap \bar{M}}_{\in \mathcal{A}}) \cup (\underbrace{M \cap \bar{N}}_{\in \mathcal{N}}) \\ &\in \tilde{\mathcal{A}} \end{aligned}$$



③ Let $\{A_n\}_{n \in \mathbb{N}} \subseteq \tilde{\mathcal{A}}$.

For each $n \in \mathbb{N}$, $A_n = A'_n \cup N_n$.

Each N_n has some $M_n \in \mathcal{A}$ with $N_n \subseteq M_n$ and $\mu(M_n) = 0$

$$\text{Then } \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} (A'_n \cup N_n) = \underbrace{\left(\bigcup_{n=1}^{\infty} A'_n \right)}_{\in \mathcal{A}} \cup \underbrace{\left(\bigcup_{n=1}^{\infty} N_n \right)}_{\in \mathcal{N}}$$

$\in \tilde{\mathcal{A}}$. $\emptyset \in \mathcal{N}$ implies $A \in \tilde{\mathcal{A}}$

(b) Let $A_1 \cup N_1 = A_2 \cup N_2$. We wish to show

$$\tilde{\mu}(A_1 \cup N_1) = \tilde{\mu}(A_2 \cup N_2) \Leftrightarrow \mu(A_1) = \mu(A_2)$$

Let $N_1 \subseteq M_1$ and $N_2 \subseteq M_2$, $M_1, M_2 \in \mathcal{A}$ and $\mu(M_1) = \mu(M_2) = 0$.

Then $A_1 \in A_2 \cup N_2 \Rightarrow A_1 \setminus A_2 \in N_2$

$$\text{So } \mu(A_1) = \mu(A_1 \cap A_2) + \underbrace{\mu(A_1 \setminus A_2)}_{\leq \mu(N_2) = 0}$$

$$= \mu(A_1 \cap A_2)$$

We can similarly show $\mu(A_2) = \mu(A_1 \cap A_2)$

(c) ① Shown

$$\text{② } \tilde{\mu}(\emptyset) = \tilde{\mu}(\emptyset \cup \emptyset) = \mu(\emptyset) = 0$$

③ Let $\{A_n\}_{n \in \mathbb{N}} \subseteq \tilde{\mathcal{A}}$ be p.d.

$$\tilde{\mu}\left(\sum_{i=1}^{\infty} A_n\right) = \tilde{\mu}\left(\sum_{i=1}^{\infty} (A_n' \cup N_n)\right)$$

$$= \tilde{\mu}\left(\left(\sum A_n'\right) \cup \left(\sum N_n\right)\right)$$

$$= \mu\left(\sum_{i=1}^{\infty} A_n'\right) = \sum_{i=1}^{\infty} \mu(A_n')$$

$$= \sum_{i=1}^{\infty} \tilde{\mu}(A_n' \cup N_n)$$

$$= \sum_{i=1}^{\infty} \tilde{\mu}(A_n)$$

(d) $\mathcal{A}$ and $\tilde{\mathcal{A}}$ are σ -algebras

μ and $\tilde{\mu}$ are measures

$$\tilde{\mu}(A) = \tilde{\mu}(A \cup \emptyset) = \mu(A) \text{ for } A \in \mathcal{A}$$

So $\tilde{\mu}$ is an extension of μ

(e) Let $\tilde{N}$ be negligible in $\tilde{\mathcal{A}}$. Then there exists $\tilde{M} \in \tilde{\mathcal{A}}$ with $\tilde{\mu}(\tilde{M}) = 0$

We have $\tilde{M} = A \cup N$ for $A \in \mathcal{A}, N \in \mathcal{N}$.

Then there exists $\tilde{M} \in \tilde{\mathcal{A}} : N \subseteq \tilde{M}$ and

$$\mu(M) = 0. \text{ Now } \tilde{\mu}(\tilde{M}) = 0 \Rightarrow \mu(A) = 0 \Rightarrow$$

$A \in \mathcal{N}. \tilde{N} \subseteq \tilde{M} \Rightarrow \tilde{N} \in \mathcal{N}. \text{ So } \tilde{N} = \emptyset \cup \tilde{N} \in \tilde{\mathcal{A}},$

$$\text{and } \tilde{\mu}(\tilde{N}) = \mu(\emptyset) = 0$$